Lecture 4: Chemical Differentiation & Magnetospheres

Planetary Systems

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Aurora australis, ISS. Credit: NASA

Learning Objectives

By the end of this lecture, you will be able to:

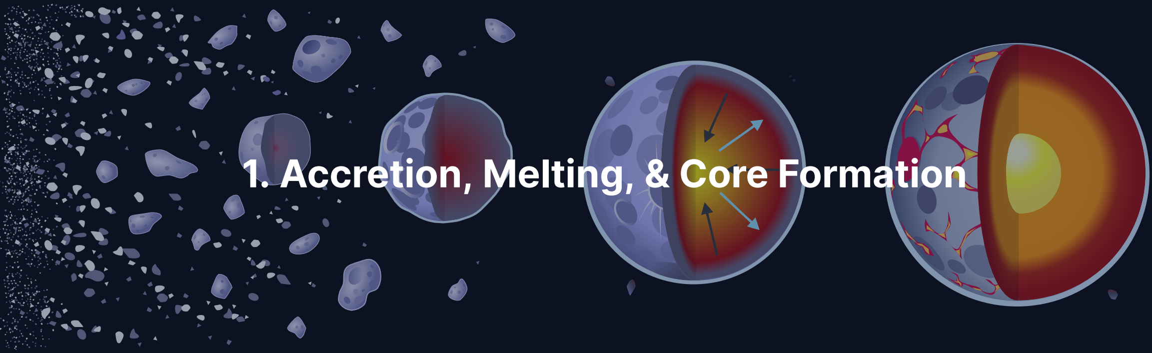
- Explain metal–silicate separation and core formation
- Use siderophile element depletion as evidence for differentiation
- Describe the Hf–W chronometer for dating core formation
- Derive the magnetic Reynolds number and assess dynamo feasibility
- Compare magnetic fields across the solar system

Recap: Lecture 3

- Four heat sources: accretional, differentiation, radioactive, tidal
- Conduction too slow for planets: $\tau \sim L^2/\kappa$
- Rayleigh number: $\mathrm{Ra} \sim 10^7$–10^8 for Earth $\Rightarrow$ vigorous convection
- Plate tectonics (mobile lid) vs. stagnant lid
- Tidal heating: Io, Enceladus, Europa

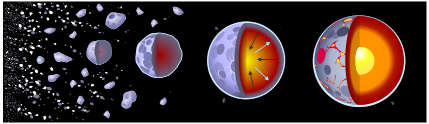
Today: The *chemical* consequences of melting, and how differentiation enables magnetic fields.

1. Accretion, Melting, & Core Formation



Planetary differentiation. Credit: Wikimedia, CC BY-SA 4.0

Metal–Silicate Separation



Credit: Wikimedia, CC BY-SA 4.0

- Accretional heating + $^{26}\mathrm{Al}$ → **magma ocean**
- Iron (ρ ~ 7000 kg m^{-3}) is immiscible with silicate (ρ ~ 3000 kg m^{-3})
- Iron droplets sink through silicate melt
- **Stokes settling velocity:**

$$v_{\mathrm{Stokes}} = \frac{2}{9}\frac{\Delta\rho\, g\, r^2}{\mu}$$

- For cm-sized droplets: v ~ 4 m s^{-1}
- Traverses magma ocean in **days**

The Magma Ocean State

After accretion, the outer few thousand kilometres of a terrestrial planet are molten.

- Depth: several hundred to >2000 km (whole-mantle for large impacts)
- Surface temperature: 2000–4000 K depending on atmosphere
- Viscosity: 10^{-1}–10^{0} Pa s (like motor oil, for molten silicate)
- Convection is **turbulent** (Rayleigh number $\gtrsim 10^{25}$)
- Cooling is set by surface radiation and greenhouse blanketing

Lifetime of a magma ocean is 10^3–10^8 yr: long enough for core formation, short compared to planetary evolution.

The Moon-Forming Giant Impact

- Mars-sized body (Theia) collided with proto-Earth ~4.5 Ga
- Re-melted most of Earth's mantle, vaporised part of it
- Created the final deep magma ocean
- Established present core–mantle structure
- Debris disk → Moon (confirmed by lunar $\varepsilon^{182}\mathrm{W}$ matching Earth's mantle)

Core formation is not a single event but a series of episodes, culminating in the giant impact.

The Late Veneer

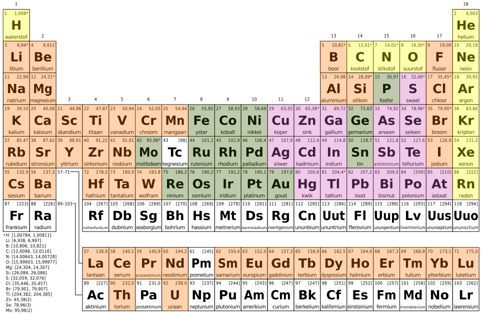
Highly siderophile elements (Ir, Pt, Os, Au) are **overabundant** in Earth's mantle compared to the deep-core equilibration prediction.

- Mantle abundances ~200× higher than expected for full metal extraction
- Relative ratios are **chondritic**, not fractionated
- Interpretation: an extra ~0.5% of Earth's mass accreted **after** core formation closed
- This "late veneer" delivered its siderophile elements to the mantle without equilibrating with the core

Source: Walker (2009); Day et al. (2016)

A smaller late veneer is also inferred for Mars and the Moon from their platinum-group element abundances.

Goldschmidt Classification



Credit: Wikimedia, CC BY-SA 3.0

Class	Affinity	Examples
Siderophile	Core (metal)	Fe, Ni, Pt, Au
Lithophile	Mantle (silicate)	Si, Mg, Al, U, T
Chalcophile	Sulfide	Cu, Zn, Pb
Atmophile	Gas	H, C, N, noble

The depletion of siderophile elements in Earth's mantle is **direct evidence** for core formation.

Metal–Silicate Partition Coefficients

How strongly does an element prefer metal vs. silicate?

$$D_i^{\mathrm{met/sil}} \equiv \frac{C_i^{\mathrm{metal}}}{C_i^{\mathrm{silicate}}}$$

Element	$D^{\mathrm{met/sil}}$	Behaviour
Ir, Os, Pt	10^4–10^6	Highly siderophile
W, Mo	10^2–10^4	Moderately siderophile
Ni, Co	10–100	Weakly siderophile
Fe	~10	Weakly siderophile
Si, Mg, O	$\ll 1$	Lithophile

Source: Rubie et al. (2015); McDonough (2003)

Earth's mantle is depleted ~1000× in Ir and Pt: direct evidence that iron-loving elements followed the metal into the core.

Why Silicon Is in the Core

Partition coefficients depend strongly on pressure, temperature, and oxygen fugacity.

- At low pressure (<10 GPa): Si is strongly lithophile, $D_{\mathrm{Si}}^{\mathrm{met/sil}} \ll 1$
- At 40–60 GPa (deep magma ocean base): $D_{\mathrm{Si}}^{\mathrm{met/sil}}$ rises to ~1
- This implies Earth's core contains **several wt%** silicon
- Si in core helps explain the ~5–10% core density deficit (Lecture 8)

Deep magma ocean partitioning (40–60 GPa) is the standard model for Earth's core composition, matching observed siderophile element depletions.

Source: Rubie et al. (2015); Fischer et al. (2015)

Hf–W Chronometry: When Did Cores Form?

- $^{182}\mathrm{Hf} \rightarrow {}^{182}\mathrm{W}$, half-life = 8.9 Myr
- Hf is **lithophile** (stays in mantle)
- W is **siderophile** (extracted to core)
- Early core formation → excess $^{182}\mathrm{W}$ in mantle

Body	ε^{182}W	**Core formation timing**
Earth	+2	30–60 Myr after solar system
Mars	+3 to +5	10–20 Myr (faster, smaller)
Moon	~Earth	Consistent with giant impact

Source: Kleine et al. (2009), *Geochim. Cosmochim. Acta*

How the Hf–W Clock Works

Two bodies diverging from chondritic composition:

Early core formation (Mars):

- Core extracts all W, stops producing $^{182}\mathrm{W}$
- Mantle keeps Hf, continues making $^{182}\mathrm{W}$
- Result: strong mantle excess (+3 to +5ε)

Late core formation (hypothetical):

- $^{182}\mathrm{Hf}$ already decayed
- No divergence between core and mantle $^{182}\mathrm{W}$
- Mantle would have $\varepsilon^{182}\mathrm{W} \approx 0$ (chondritic)

Earth's moderate excess (+2ε) places core formation in the 30–60 Myr window.

Magma Ocean Crystallisation

Core formation is not the end. As the magma ocean cools:

- Crystallisation starts at the **base** (high pressure raises melting point)
- Solidification progresses **upward**
- First minerals: **bridgmanite** ($MgSiO_3$) in the lower mantle
- Then **olivine** ($(Mg, Fe)_2SiO_4$) at shallower depths
- Final residual melt at the top: enriched in incompatible elements

Source: Elkins-Tanton (2012), *Annu. Rev. Earth Planet. Sci.*

Incompatible Element Enrichment

- Large ions (U^{4+}, Th^{4+}, K^{+}) do not fit well into crystal lattices
- Progressively concentrated in the shrinking residual liquid
- Eventually trapped in the **crust**
- Continental crust is ~50–100× enriched in U, Th, K vs. upper mantle
- Roughly **half** of Earth's radiogenic heat is produced in the crust, even though the crust is <1% of Earth's mass

Incompatible element enrichment is why Earth's crust is both radioactive (heat source) and geochemically distinct from the mantle below.

Mantle Reservoirs

Magma ocean crystallisation and billions of years of processing leave distinct mantle reservoirs:

- DMM **Depleted MORB Mantle**: upper mantle source of mid-ocean ridge basalts; depleted by partial melting and crustal extraction.
- EM/OIB **Enriched Mantle / Ocean Island Basalt source**: deep plumes sample reservoirs with preserved incompatible elements; may represent primordial mantle or recycled oceanic crust.
- HIMU "High-μ" reservoir: high U/Pb, likely very old subducted oceanic crust.

Isotopic diversity of ocean island basalts (Hawaii, Iceland, Samoa) is direct evidence for long-lived mantle heterogeneity.

The Lunar Magma Ocean: Direct Evidence

The Moon shows the clearest record of magma ocean differentiation:

- **Anorthositic highlands:** low-density plagioclase feldspar floated to the top of the crystallising lunar magma ocean ~4.4 Ga
- Apollo samples of anorthosite date to 4.36–4.44 Ga (within 200 Myr of solar system formation)
- **KREEP** layer: the very last dregs of crystallisation, concentrated in **K**, **R**are **E**arth **E**lements, and **P**hosphorus
- Found at shallow depths under the nearside maria (Procellarum KREEP Terrane)
- Lunar crust thickness varies from ~30 km (nearside) to ~70 km (farside)

The Moon is the textbook case of magma ocean differentiation; its exposed anorthosite confirms the theoretical crystallisation sequence.

Volatile Delivery: Classical vs. Modern View

Classical picture:

- Inner solar system forms dry, inside the water snow line
- Volatiles delivered **late** by comets and outer-belt asteroids

Revised picture (post-2010):

- All chondrite groups accreted some water ice
- Snow line migration + Jupiter's growth created an isotopic bifurcation
- Most of Earth's water was already present at accretion, not delivered afterwards

Giant impacts can either deliver **or** strip volatiles depending on impact geometry and atmospheric state.

Outgassing from the Magma Ocean

During magma ocean cooling, dissolved volatiles **outgas** into the overlying atmosphere.

Speciation depends on **oxygen fugacity** (f_{O_2}), set by the mantle redox state:

Reducing (low f_{O_2}): H_2, CO, N_2, CH_4

Oxidising (high f_{O_2}): H_2O, CO_2, N_2, SO_2

The oxidation state of the mantle at magma ocean solidification sets the initial atmosphere, which then conditions habitability (Lectures 5–6).

Water in Silicate Melt: A Square-Root Law

The solubility of water in silicate magma follows:

$$X_{\mathrm{H_2O}} \propto p_{\mathrm{H_2O}}^{1/2}$$

- $X_{\mathrm{H_2O}}$ = mole fraction dissolved in melt
- $p_{\mathrm{H_2O}}$ = partial pressure of water vapour above melt
- Higher atmospheric $p_{\mathrm{H_2O}}$ → more water retained in the magma
- Self-limiting feedback during atmosphere build-up

Consequence: a planet cannot outgas its entire water budget; some fraction remains locked in the solidifying mantle, potentially for Gyr.

Source: Hirschmann (2012), *Earth Planet. Sci. Lett.*

Meteoritic Dichotomy: NC vs. CC

Meteorites split into two isotopic groups: **non-carbonaceous (NC)** and **carbonaceous (CC)**.

- NC and CC have distinct $^{50}\mathrm{Ti}$, $^{54}\mathrm{Cr}$, $^{48}\mathrm{Ca}$ ratios
- They reflect two separate reservoirs in the early solar nebula
- Interpretation: **Jupiter** formed early, blocking mixing between inner and outer disk
- Both reservoirs contributed to Earth's accretion, but in changing proportions over time

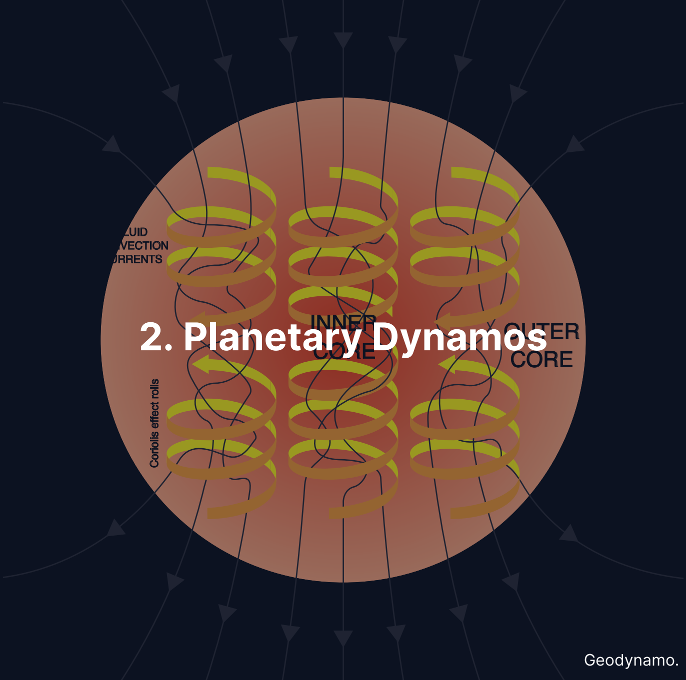
Source: Kruijer et al. (2017); Lichtenberg et al. (2021)

This dichotomy is one of the strongest meteoritic constraints on the timing of giant planet formation.

Break

Earthrise, Apollo 8 (1968). Credit: NASA

2. Planetary Dynamos



Geodynamo. Credit: Wikimedia, CC BY-SA 4.0

Magnetic Fields and Habitability

A magnetic field is more than a compass-maker. It shapes a planet's long-term viability.

- Deflects the solar wind ⇒ reduces ion sputtering of the upper atmosphere
- Limits cosmic ray flux to the surface ⇒ lower radiation dose
- Protects the ozone layer from charged-particle destruction
- On Venus, the lack of a field is one likely contributor to hydrogen escape
- On Mars, the loss of the field preceded the loss of ~0.8 bar of atmosphere (MAVEN)
- Debated: how essential is a global field for a habitable surface?

Source: Lammer et al. (2003); Jakosky et al. (2018)

Why Magnetic Fields Matter

- **Atmospheric shielding:** deflects solar wind, reduces erosion
  Mars lost its field ~4 Ga → lost most of its atmosphere
- **Surface habitability:** reduces radiation dose at the surface
- **Geological record:** magnetic minerals record field history (paleomagnetism)
- **Interior probe:** field properties constrain the deep interior

The Induction Equation

Magnetic fields in a conducting fluid obey:

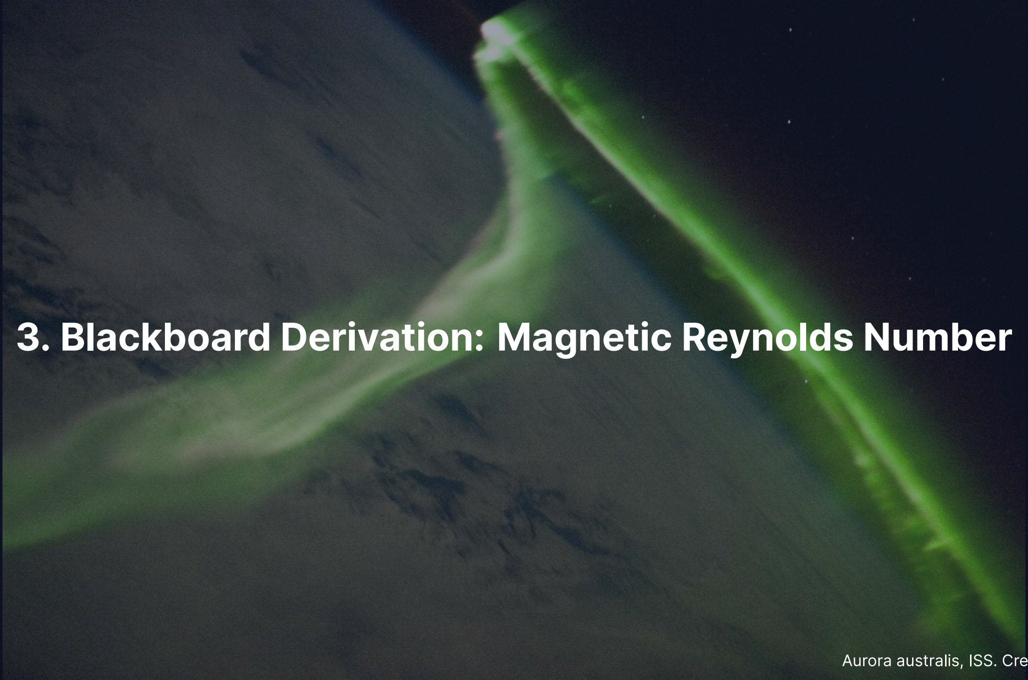
$$\frac{\partial \vec{B}}{\partial t} = \underbrace{\nabla \times (\vec{v} \times \vec{B})}_{\text{advection}} + \underbrace{\eta \nabla^2 \vec{B}}_{\text{diffusion}}$$

- **Advection:** fluid motions amplify the field (create flux)
- **Diffusion:** ohmic resistance destroys the field (decay)
- $\eta = 1/(\mu_0 \sigma)$ = magnetic diffusivity

Which term wins determines whether a dynamo can operate.

Three Requirements for a Dynamo

1. **Electrically conducting fluid**
   Terrestrial planets: liquid iron alloy in the core
   Gas giants: metallic hydrogen
2. **Convection**
   Thermal buoyancy (core hotter than mantle)
   Compositional buoyancy (inner core crystallisation releases light elements)
3. **Sufficient flow vigour**
   Advection must dominate over diffusion
   Quantified by the **magnetic Reynolds number** $\mathrm{Rm} \gtrsim 10\text{–}100$



3. Blackboard Derivation: Magnetic Reynolds Number

Aurora australis, ISS. Credit: NASA

Deriving the Magnetic Reynolds Number

Goal: From the induction equation, derive $\mathrm{Rm} = UL/\eta$ and show that advection dominates in Earth's core.

Dimensional analysis of each term:

Advection: $|\nabla \times (\vec{v} \times \vec{B})| \sim UB/L$

Diffusion: $|\eta \nabla^2 \vec{B}| \sim \eta B/L^2$

The Magnetic Reynolds Number

$$\mathrm{Rm} = \frac{\text{advection}}{\text{diffusion}} = \frac{UB/L}{\eta B/L^2}$$

The field strength B **cancels**:

$$\boxed{\mathrm{Rm} = \frac{UL}{\eta}}$$

- $\mathrm{Rm} \gg 1$: advection dominates, field “frozen in” to the flow
- $\mathrm{Rm} \ll 1$: diffusion wins, field decays
- Critical value: $\mathrm{Rm}_c \sim 10$–100

Key insight: Rm depends on the **flow**, not the field.

Application: Earth's Outer Core

Parameter	Symbol	Value
Core radius	L	3.5×10^6 m
Flow velocity	U	5×10^{-4} m s^{-1}
Magnetic diffusivity	η	~1 m^2 s^{-1}

$$\mathrm{Rm}_\oplus = \frac{5 \times 10^{-4} \times 3.5 \times 10^6}{1} \approx 1750$$

$\mathrm{Rm} \approx 1750 \gg \mathrm{Rm}_c$: advection overwhelmingly dominates.

Ohmic diffusion timescale: $\tau_{\mathrm{ohm}} \sim L^2/\eta \approx 400{,}000$ yr $\ll$ Earth's age $\Rightarrow$ field must be **continuously regenerated**.

Dynamo Scaling Laws

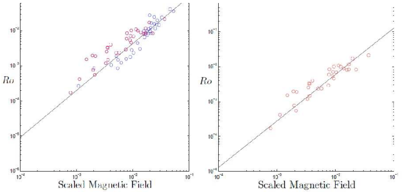


Figure 2. Computed values of Ro versus $B_{rms}/\Omega D\sqrt{\rho\mu}$ taken from the data set of Christensen & Aubert (2006). (a) The full data set, $0.06 < \mathrm{Pr}_m < 10$, with $\mathrm{Pr}_m < 1$ coloured red and $\mathrm{Pr}_m > 1$ coloured blue. (b) A restricted data set, $0.06 < \mathrm{Pr}_m < 1$. Power-law (28) is shown for comparison.

If one has sufficient confidence in power-law (28), it is tempting to use it to estimate Ro, and hence $P/\Omega^3\ell_{//}^2$, in the core of the planets, although this is a rather substantial extrapolation. In any event, the various estimates of $B_z/\Omega R_C\sqrt{\rho\mu}$ are tabulated in table 2 and they are surprisingly consistent across the different planetary dynamos. Expression (28) then suggests Rossby numbers in the range $\mathrm{Ro} \sim 10^{-8} \rightarrow 10^{-6}$, as shown

Credit: Davidson (2013), arXiv:1302.7140, after Christensen & Aubert (2006)

Christensen & Aubert (2006) showed that dynamo simulations collapse onto simple power laws:

- Field strength ∝ (convective power)$^{1/3}$
- **Independent of rotation rate**
- Predicts Earth, Jupiter, Mercury, Ganymede from first principles
- Works across 2 orders of magnitude in parameters

Planetary magnetic field strength is set by the **available convective**

Hf–W Chronometry: Beyond the First Estimate

Modern refinements of the core formation timescale.

- Pressure-dependent partitioning: $D_W^{\mathrm{met/sil}}$ decreases at high P
- At deep magma ocean pressures (~60 GPa), D_W is ~10 instead of ~40 (surface)
- This **delays** the inferred Earth core formation time from 30 Myr to 60–100 Myr
- Complementary isotope systems (Pd–Ag, Pt–Os) cross-check the timing
- Evidence for **disequilibrium core formation**: not all metal re-equilibrates with silicate before sinking
- Earth's core likely formed in episodic accretion events, not continuous

Source: Rudge et al. (2010); Kleine et al. (2017)

The Archean-Proterozoic Paleomagnetic Record

Geologists trace Earth's dynamo history through preserved magnetic signals.

- **Archean** (>2.5 Ga): paleomagnetic records in komatiites and banded iron formations
- Field has been **continuous** from ≳3.4 Ga to today (confirmed at multiple sites)
- **Proterozoic** (2.5–0.54 Ga): Apparent Polar Wander (APW) paths reveal plate motions and supercontinent cycles
- Paleointensity from Thellier experiments: early field strength similar to or slightly stronger than modern
- Reversal statistics: Archean reversal rate poorly constrained; modern rate set in the last ~500 Myr
- **Singularity puzzle**: some Proterozoic intervals show very frequent reversals followed by superchrons

Source: Biggin et al. (2015); Tarduno et al. (2020)

How Old Is Earth's Magnetic Field?

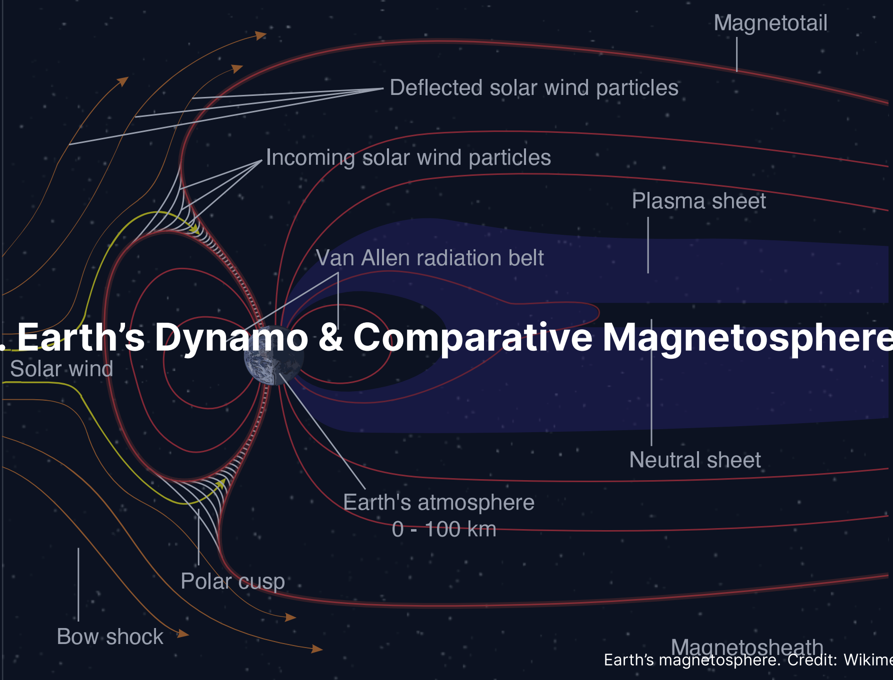
Paleomagnetism records the field in ancient rocks.

- Oldest confirmed field: ~**3.4–3.5 Gyr** (Archean South Africa, Tarduno 2010)
- Single-crystal zircon paleomagnetism pushes this to **4.2 Gyr** (Tarduno et al. 2015)
- Implies Earth's dynamo has been continuously active for most of its history
- Constrains the core cooling rate and the earliest habitability of the surface

The dynamo was active during the Hadean eon, shielding Earth's atmosphere during the time when life was emerging.

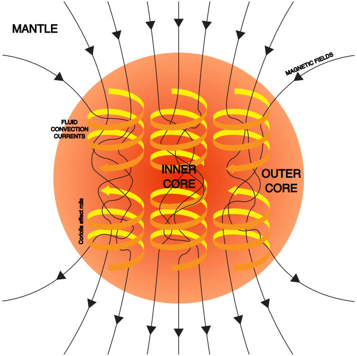
Source: Tarduno et al. (2010, 2015), *Science*

4. Earth’s Dynamo & Comparative Magnetospheres



Earth’s magnetosphere. Credit: Wikimedia, CC BY-SA 3.0

Earth's Geodynamo



Credit: Wikimedia, CC BY-SA 4.0

- **Outer core:** liquid Fe alloy (1220–3480 km from centre)
- Two driving mechanisms:
    1. Thermal convection (core hotter than mantle)
    2. Compositional convection (inner core crystallisation)
- Predominantly **dipolar** field
- Tilted ~11° from rotation axis
- Surface field: 25–65 μT

Earth's Core: Composition & Growth

Inner core (0–1220 km):

- Solid iron–nickel alloy
- Growing at ~0.5 mm/yr
- Age: 0.5–1.5 Gyr (not as old as Earth)
- Latent heat of crystallisation powers the dynamo today

Outer core (1220–3480 km):

- Liquid Fe–Ni alloy + light elements
- ~5–10% less dense than pure Fe
- Light elements: S, Si, O, C, H
- Vigorous convection ⇒ dynamo

Compositional convection (inner core crystallisation ejects light elements) is the **dominant** driver of today's geodynamo.

Secular Variation of the Field

Earth's field is not static. It varies on many timescales:

- **Secular variation** (decades to centuries): non-dipole components grow, decay, drift
- **Westward drift**: non-dipole features migrate westward at ~0.2°/yr, implying differential rotation between core and mantle
- **South Atlantic Anomaly (SAA)**: a ~30% reduction in field strength over South America and the South Atlantic
- SAA is the surface expression of a reversed-flux patch at the core–mantle boundary
- Expanding and weakening over the past century; potential precursor to a reversal (but unconfirmed)

Source: Finlay et al. (2020), Swarm satellite constellation data

Magnetic Field Reversals

Credit: Wikimedia, CC BY-SA 3.0

- N and S magnetic poles **swap**
- Current rate: ~4–5 per Myr
- Each reversal takes ~1000–10,000 yr
- Field weakens, becomes multipolar, re-establishes
- **Magnetic stripes** on ocean floor: key evidence for seafloor spreading

The Cretaceous Normal Superchron

Not all times have frequent reversals.

- **Cretaceous Normal Superchron** (CNS): ~124–84 Ma
- The field kept a **single polarity for 40 Myr**
- Also a Permo-Carboniferous reversed superchron (~312–262 Ma)
- Superchrons may reflect changes in core–mantle boundary heat flow, controlled by slow mantle convection patterns

Reversal rate is not constant; it is modulated by the thermal evolution of the core–mantle boundary over hundreds of millions of years.

Source: Biggin et al. (2012); Olson & Amit (2015)

Magnetic Fields Across the Solar System

Body	Field type	Surface field	Rel. dipole
Earth	Active dynamo	25–65 μT	1
Mercury	Active (weak)	~0.3 μT	5×10^{-4}
Venus	None detected	<0.01 μT	$< 10^{-5}$
Mars	Remnant crustal	up to 1500 nT	n/a
Jupiter	Active (strong)	400–1400 μT	20,000
Saturn	Active	~20 μT	~600
Ganymede	Active	~0.7 μT	1.5×10^{-3}

Jupiter's field is ~20,000× Earth's, generated in metallic hydrogen.

The Venus Magnetic Field Puzzle

Venus: Earth's twin in size and composition, yet **no magnetic field**.

- Slow rotation (243-day period)? Simulations say this alone is insufficient.
- **No inner core?** Without compositional convection, thermal convection alone may be too weak.
- **Stagnant lid tectonics?** Mantle may not extract core heat efficiently enough.

Most likely: **combination** of reduced core cooling (no plate tectonics) and possibly no inner core nucleation.

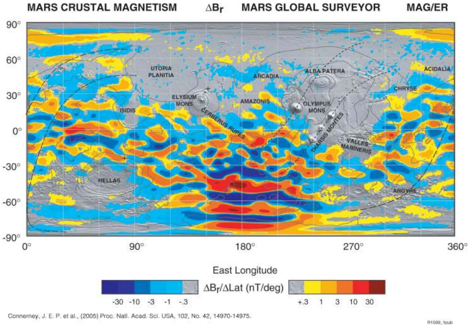
Mercury: An Offset Dynamo

Mercury's magnetic field is weak but intriguing.

- Surface field: ~0.3 μT, 1% of Earth's
- Field is **offset ~480 km north** of the geometric centre
- Dominated by the dipole component (g_{10}), weak higher harmonics
- MESSENGER (2011–2015) mapped this asymmetry
- Interpretation: thin liquid outer shell above a partially solidified deeper core
- Limits convective vigour and field strength

Source: Anderson et al. (2012), *Journal of Geophysical Research*

Mars: A Fossil Magnetic Field



Credit: NASA/GSFC

- No global field today
- Intense **crustal magnetisation** in southern highlands
- Absent in young northern lowlands and large basins

This tells us:

1. Mars **had** an active dynamo
2. Dynamo died ~4.1 Ga
3. Atmosphere left unshielded → erosion by solar wind

Mars Dynamo: A More Recent Shutdown?

The classical Acuña (1999) interpretation placed the dynamo death at ~4.1 Ga. Recent work has revised this:

- Mittelholz et al. (2020): magnetisation in the 3.9 Ga Lucus Planum lavas
- Implies Mars dynamo was **intermittent** or shut down **later** than thought
- Consistent with core cooling becoming insufficient over 4.1–3.9 Ga
- MAVEN: measured present-day atmospheric loss rate of ~100 g/s (O^+)
- Extrapolated over 4 Gyr: consistent with Mars losing $\gtrsim$ 0.8 bar of CO_2

Source: Mittelholz et al. (2020), *Science Advances*; Jakosky et al. (2018)

Jupiter: Dynamo in Metallic Hydrogen

Jupiter has the strongest magnetic field in the solar system, ~20,000× Earth's dipole moment.

- Molecular H_2 becomes **metallic** above ~100 GPa
- Transition at ~0.85 Jupiter radii from the centre
- Metallic H has conductivity comparable to liquid metals
- Dynamo operates in the deep metallic-H layer
- Juno gravity measurements show the transition is **gradual**, not sharp

Juno has revealed a complex, asymmetric surface field with a "Great Blue Spot" (reversed-flux patch in the northern hemisphere).

Source: Connerney et al. (2022); Bolton et al. (2017)

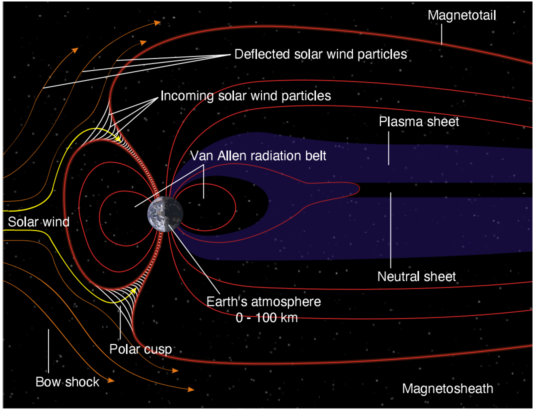
Ganymede: The Only Moon with a Dynamo

Ganymede is the **only moon** in the solar system with an intrinsic magnetic field.

- Moon discovered by Galileo Galilei (1610); intrinsic field detected by NASA *Galileo* spacecraft (1996): surface field ~0.7 μT
- Dipole tilted ~176° (nearly anti-parallel to Jupiter's field)
- Surprising: Ganymede is small (R = 2634 km), should be thermally dead
- Possible explanations:
  - Tidal heating from Laplace resonance sustains core convection
  - Slow freezing of the outer core releases compositional buoyancy
- JUICE (arriving 2031) will map Ganymede's field in detail

Source: Kivelson et al. (1996); Bland et al. (2008)

Magnetosphere–Solar Wind Interaction



Credit: Wikimedia, CC BY-SA 3.0

- **Bow shock:** solar wind decelerated from supersonic to subsonic
- **Magnetopause:** pressure balance
  $\frac{1}{2}\rho_{\mathrm{sw}} v_{\mathrm{sw}}^2 = \frac{B^2}{2\mu_0}$
  Standoff: ~10 $R_\oplus$
- **Magnetotail:** >200 $R_\oplus$
- **Plasmasphere:** cool dense plasma on closed field lines

Magnetopause Standoff Distance

Pressure balance at the magnetopause:

$$\underbrace{\frac{1}{2}\rho_{\mathrm{sw}} v_{\mathrm{sw}}^2}_{\text{solar wind ram pressure}} = \underbrace{\frac{B_{\mathrm{mp}}^2}{2\mu_0}}_{\text{magnetic pressure}}$$

For a dipole field, $B \propto r^{-3}$, so the standoff distance scales as:

$$r_{\mathrm{mp}} \propto \left(\frac{B_0^2}{\mu_0 \rho_{\mathrm{sw}} v_{\mathrm{sw}}^2}\right)^{1/6}$$

- Earth under typical solar wind: $r_{\mathrm{mp}} \approx 10\, R_\oplus$
- During a CME: compressed to ~6 $R_\oplus$
- Jupiter (much stronger field): ~50–100 R_J

Magnetotail Reconnection

- On the night side, Earth's field is stretched into a long tail (>200 $R_\oplus$)
- Opposing field lines in the magnetotail can **reconnect**, releasing stored magnetic energy
- Reconnection accelerates particles toward both Earth and downstream
- Earthward-moving particles precipitate into the polar atmosphere → **aurorae**
- Downstream particles are ejected as **plasmoids**
- Reconnection drives **magnetospheric substorms** observed every few hours during active solar conditions

Magnetic reconnection is the engine of space weather, converting solar wind kinetic energy into particle acceleration and auroral emission.

The Van Allen Radiation Belts

Earth's magnetosphere traps energetic particles in two toroidal belts:

Inner belt ~1.5 $R_\oplus$: high-energy protons (10–100 MeV), produced by cosmic-ray interaction with the atmosphere (CRAND process)

Outer belt ~4–5 $R_\oplus$: high-energy electrons (0.1–10 MeV), injected during geomagnetic storms and accelerated by wave–particle interactions

- Discovered by Explorer 1 (1958)
- Major hazard to spacecraft (electronics) and crewed missions
- Van Allen Probes (2012–2019): discovered a third, transient belt

Source: de Pater & Lissauer (2010); Baker et al. (2013)

Aurorae: Light from the Magnetosphere

Credit: NASA/ISS

Solar wind particles channelled along field lines to polar regions:

- **Green:** excited O at 100–200 km (557.7 nm)
- **Red:** excited O at 200–400 km (630.0 nm)
- **Blue/violet:** excited N_2 bands

Observed on Jupiter, Saturn, and Uranus: any magnetised planet.

Why Aurorae Happen at the Poles

- Earth's dipole focuses incoming particles toward the magnetic poles
- **Cusps** near the poles are regions where solar wind has direct access to the atmosphere
- Reconnection in the magnetotail releases stored energy
- Accelerated electrons precipitate down field lines into the atmosphere
- Energies of 1–100 keV; penetrate to 100–400 km altitude
- Emission lines and bands are fingerprints of the excited species

Aurorae map the coupling between the solar wind and the magnetosphere, lit up by atmospheric atoms.

Jupiter’s Aurorae: The Most Powerful in the Solar System

- Total UV power: 10^{13}–10^{14} W (10^3–10^4× Earth’s auroral power)
- Powered by:
  - Jupiter’s rapid rotation (~10 hr)
  - Mass loading from Io’s volcanoes (~1 tonne/s of sulfur and oxygen)
  - Solar wind interaction (secondary)
- Io, Europa, and Ganymede each create **auroral footprints** by electromagnetic coupling
- Observed in UV by HST, Juno-UVS; infrared from ground-based telescopes

Source: Connerney et al. (2022); Mauk et al. (2017) on Juno plasma observations

5. Recent Advances & Outlook

Aurora australis. Credit: NASA/ISS

Juno: Jupiter's Asymmetric Dynamo

- Juno (2016–) has measured Jupiter's gravity and magnetic field in unprecedented detail
- Surface field dramatically **non-dipolar**
- Northern hemisphere: intense concentrated "flux lobe"
- "Great Blue Spot": a patch of reversed polarity near the equator
- Magnetic equator offset from the rotational equator
- Implies dynamo operates at relatively shallow depths in the stably-stratified dilute core (Lecture 8)

Source: Connerney et al. (2022), *Journal of Geophysical Research*

BepiColombo: New Eyes on Mercury

- ESA/JAXA joint mission, launched 2018
- Multiple Mercury flybys (2021–2025) en route to orbit insertion
- Already confirmed the weak but active dipolar field
- Revealed sodium ion cyclotron waves in Mercury's magnetosphere
- Will map the offset field with unprecedented precision from polar orbit
- Two orbiters: Mercury Planetary Orbiter (MPO) + Mercury Magnetospheric Orbiter (MMO)

Source: Benkhoff et al. (2021), *Space Science Reviews*

JUICE: Mapping Ganymede's Dynamo

- Jupiter Icy Moons Explorer, launched April 2023
- Arrives at Jupiter in 2031; Ganymede orbit insertion 2034
- First orbiter of a moon other than Earth's
- Will map Ganymede's intrinsic field at high resolution
- Separate intrinsic field from induced field (ocean signature)
- Tests the hypothesis that Ganymede has a **subsurface ocean**
- Bridges Lecture 4 (dynamo physics) and Lecture 8 (interiors)

Source: Grasset et al. (2013) on JUICE science goals

MAVEN and the Escape of Mars' Atmosphere

- NASA MAVEN (2014–) measures Mars atmospheric loss in situ
- Total escape rate: ~1–2 kg/s of all species (O, H, CO_2)
- Dominated by photochemical escape of oxygen
- Escape accelerates during solar storms
- Extrapolated over 4 Gyr: ~0.8 bar of CO_2 lost, consistent with a warmer early Mars
- **Directly ties** the loss of the dynamo (Lecture 4) to the loss of the atmosphere (Lecture 5)

Source: Jakosky et al. (2018), *Icarus*; Lillis et al. (2017)

Psyche: Visiting a Naked Core

NASA Psyche launched in October 2023, arrives at asteroid 16 Psyche in 2029.

- Target: a ~220 km asteroid, one of the largest M-type
- Long suspected to be the exposed metallic core of a differentiated planetesimal
- Bulk density ~4 g/cm^3 (higher than stony, lower than pure iron)
- Instruments: multispectral imager, magnetometer, gamma-ray spectrometer
- Will test whether Psyche has remnant magnetisation (ancient dynamo relic)
- A direct sample of the kind of body L4's theory describes

Source: Elkins-Tanton et al. (2017), *J. Geophys. Res. Planets*

Open Questions in Planetary Magnetism

Despite decades of work, key questions remain:

- **When did the inner core nucleate?** (Estimates range from 0.5 to 2 Gyr ago)
- **How did Earth sustain a dynamo before inner core formation?**
- **Why does Venus have no detectable field?** Inner core, rotation, or tectonics?
- **Why is Mercury's field so weak yet active?** What powers it?
- **What drives Ganymede's dynamo?** (A moon should be thermally dead)
- **How do Uranus and Neptune generate multipolar fields?** (Exotic dynamo physics)
- **Will Earth reverse soon?** (No reliable prediction method exists)
- **How important are magnetic fields for habitability?** (Crucial or secondary?)

Future missions (JUICE, BepiColombo, Europa Clipper, MSR) will address several of these

Summary

1. Magma oceans drive **metal–silicate separation**: iron sinks, silicates float
2. Hf–W chronometry dates Earth's core formation to ~30–60 Myr (classic); refined to ~60–100 Myr with pressure-dependent partitioning
3. Magma ocean crystallisation creates chemically distinct layers (crust, mantle reservoirs)
4. Dynamos require: conducting fluid + convection + sufficient Rm ($\gtrsim$10–100)
5. Earth's $\mathrm{Rm} \approx 1750$: vigorous dynamo action
6. Mars lost its dynamo around 4.1–3.9 Ga (possibly intermittent, Mittelholz 2020) $\rightarrow$ atmospheric erosion
7. Venus has no detectable field: likely a combination of stagnant-lid tectonics, slow rotation, and/or absence of an inner core

Questions?

Next: **Lecture 5. Atmospheres I: Composition, Structure, & Energy Balance**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`